

4.2 Power Series Worksheet

1. Determine the Taylor series centered at 9 for $f(x) = e^{-2x}$.

2. Determine the Taylor series centered at 1 for $f(x) = x^3$.

3. Determine the Maclaurin series centered at 0 for $f(x) = \cos(2x)$

4. Determine the interval and radius of convergence for the series $\sum_{n=1}^{\infty} n(x+5)^{n+1}$.

5. Find the interval and radius of convergence for the power series $\sum_{k=0}^{\infty} \left(\frac{9}{4}\right)^k x^{2k}$.

6. Find the interval and radius of convergence for the power series $\sum_{k=1}^{\infty} \frac{(x-2)^k}{k}$.

7. Find the interval and radius of convergence for the power series $1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \dots$